

PDAC-BioPortal: System Architecture

1. PUBLIC TRANSCRIPTOMIC DATASETS

TCGA-PAAD vs GTEx Bulk RNA-seq

- Bulk RNA-seq: 178 tumors vs 171 normal reference
- Tumor vs Normal: Differential expression (DEGs)
- Concordance: Wilcoxon rank-sum & limma-voom
- Reprocessing: UCSC Xena Toil unified pipeline

GSE225767 (SBRT) Bulk RNA-seq

- Neoadjuvant SBRT: 55 bulk RNA-seq samples
- Study Design: 26 pre-tx FNA vs 29 post-tx resection
- Response Analysis: Responders vs Non-responders
- Radioresistance: DEGs and pathway alterations

GSE202051 snRNA-seq

- Single-nucleus RNA-seq: 43 clinical specimens
- Atlas Scale: 224,988 profiled single nuclei
- Cell Lineages: Ductal, acinar, stroma & immune
- Cell Atlas: Stratified 20k cell fast interactive viewer

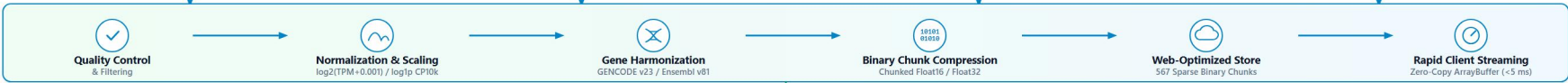
GSE274103 Spatial Visium

- Spatial Transcriptomics: 10x Visium array
- Cohort Scale: 5 patients (23,436 spatial spots)
- Histology Alignment: Tissue-level H&E co-registration
- Microenvironment: Tumor-stroma niche patterns

DATA TYPES

- Bulk RNA-seq (Tumor / Normal / SBRT)
- Single-nucleus RNA-seq (snRNA-seq)
- Spatial Transcriptomics (10x Visium)

2. DATA PROCESSING & STORAGE



3. PDAC-BioPortal (Next.js Web Interface) CORE ANALYTICAL PLATFORM

Integrated Exploration Across Five Analytical Modules

TCGA/GTEx Explorer

- DEGs & Volcano Plots: Interactive Volcano & MA plots
- Expression Profiling: Heatmaps & multi-gene boxplots with log2(TPM) violin plots
- Correlation Analysis: Spearman / Pearson correlation with strict cohort isolation
- Statistical Testing: Empirical Bayes limma-voom & Benjamini-Hochberg FDR control
- Tumor vs Normal Contrast: Normal pancreas baseline resolution accounting for acinar dominance

SBRT Explorer

- Pre/Post SBRT DEGs: Pre-treatment vs post-resection
- Treatment Response: Pathologic regression grading (Responders vs Non-responders)
- Heatmaps & Correlation: Hierarchically clustered gene sets and co-expression networks
- Radioresistance Signatures: NRF2 antioxidant, DNA repair & hypoxic adaptation markers
- Unpaired Cohort Model: Rigorous statistical evaluation of independent patient cohorts

Single-nucleus Atlas

- UMAP Visualization: High-speed 2D cell projections
- Cell Type Clustering: Ductal 1/2, acinar, PSC, fibroblasts, endothelial, & immune subsets
- Annotation & Marker Genes: Continuous expression overlays & single-nucleus distribution plots
- Cell Lineage Exploration: Acinar-to-ductal metaplasia (ADM) & malignant epithelial transitions
- Subcluster Profiling: Stratified 20k nuclei sampling with accurate proportion calculations

Spatial Explorer

- Spot-Level Expression: Continuous plasma gradient maps
- H&E Image Overlay: Full-resolution histological image co-registration & alpha blending
- Spatial Neighborhoods: Tumor-stroma microenvironment niches & spatial clustering
- Real-Time HiX Testing: Instantaneous hover inspection & spot-level count reconstruction
- Multi-Patient Pancreas: 5 PDAC patient tumor sections spanning 23,436 spatial spots

Functional / Pathway Analysis

- ORA (Over-Representation): Hypergeometric testing on DEGs
- GSEA (Gene Set Enrichment): Pre-ranked enrichment scores & leading-edge gene identification
- Pathway Libraries: MSigDB Hallmark, KEGG & Reactome oncology & metabolic signatures
- Cross-Dataset Enrichment: Synchronized pathway exploration across bulk, single-cell & spatial
- Interactive Visualizers: Enrichment bubble charts, bar plots & pathway overlap networks

Unified Data Access & Cross-Dataset Integration

- Consistent Gene Annotation: Standardized Ensembl ID / HGNC gene symbol resolver with automatic alias resolution
- Harmonized Metadata: Standardized clinical annotations (TNM stage, survival status, histologic grade, response)
- Synchronized Visualization & Querying: One-click cross-module sync across TCGA, SBRT, snRNA, and Spatial tabs

5. USER VERIFICATION (Human-in-the-Loop)

Cross-checking & Verification

Users validate AI responses by cross-checking across interactive portal visualizations and underlying data. Ensures transparency, accuracy, and trust.

- Direct verification against Volcano plots, Heatmaps, Boxplots, UMAP cell clusters, and Visium spatial overlays
- Dual human-in-the-loop validation loop provides full empirical traceability for scientific manuscript reporting

4. PDACopilot AI Assistant (Decoupled Grounded Layer)

User Query & Active Context

- Natural language question paired with live portal state
- Captures active dataset, selected gene, filters, and cohort contrast

Structured Context Retrieval

- Deterministic retrieval of relevant tables, metadata, and DEG results
- Zero-latency extraction of exact log2FC, FDR values, and cell proportions

PDACopilot Reasoning Engine

- Google Gemini 2.0 Flash (or pluggable LLaMA 3.3 / GPT-4o)
- Grounded analytical reasoning strictly over retrieved context only
- Guardrail locks prevent ungrounded extrapolation or hallucination

Evidence-Tagged Response

- Bioinformatic answer synthesized with exact numeric citations
- Evidence checklist verification tags and confidence indicators

Reproducible Summary Export

- Export answers, retrieved evidence tables, and query audit trail
- Ensures full manuscript transparency and scientific reproducibility

DESIGN PRINCIPLES

- Modular & Scalable
- Transparent & Evidence-Driven
- Data Integrity & Reproducibility
- Decoupled AI for Reliability

